

Scenario 1

Question: Display Employee ID, Employee Name, Department Name, Salary for employees with valid department. Sort by Salary DESC.

MySQL Answer:

```
SELECT
    e.Emp_ID,
    e.Emp_Name,
    d.Department_Name,
    e.Salary
FROM Employee e
INNER JOIN Department d
    ON e.Department_ID = d.Department_ID
ORDER BY e.Salary DESC;
```

Scenario 2

Question: Customers who have not placed orders. Show Customer ID, Name, City, Order ID. Sort by Name.

MySQL Answer:

```
SELECT
    c.Customer_ID,
    c.Customer_Name,
    c.City,
    o.Order_ID
FROM Customer c
LEFT JOIN Orders o
    ON c.Customer_ID = o.Customer_ID
WHERE o.Order_ID IS NULL
ORDER BY c.Customer_Name ASC;
```

Scenario 3

Question: Departments without employees.

MySQL Answer:

```
SELECT
    d.Department_ID,
    d.Department_Name,
    e.Emp_ID,
    e.Emp_Name
FROM Department d
LEFT JOIN Employee e
    ON d.Department_ID = e.Department_ID
WHERE e.Emp_ID IS NULL
ORDER BY d.Department_Name;
```

Scenario 4

Question: Products in both, only master, only warehouse.

MySQL Answer:

```
SELECT
    pm.Product_ID,
    pm.Product_Name,
    ws.Quantity,
    CASE
```

```

        WHEN pm.Product_ID IS NOT NULL AND ws.Product_ID IS NOT NULL THEN
'Both'
        WHEN pm.Product_ID IS NOT NULL THEN 'Only Product_Master'
        ELSE 'Only Warehouse_Stock'
    END AS Status
FROM Product_Master pm
LEFT JOIN Warehouse_Stock ws
    ON pm.Product_ID = ws.Product_ID
UNION
SELECT
    ws.Product_ID,
    NULL,
    ws.Quantity,
    'Only Warehouse_Stock'
FROM Warehouse_Stock ws
LEFT JOIN Product_Master pm
    ON ws.Product_ID = pm.Product_ID
WHERE pm.Product_ID IS NULL
ORDER BY Product_ID;

```

Scenario 5

Question: Employee reporting hierarchy.

MySQL Answer:

```

SELECT
    e.Emp_Name AS Employee_Name,
    m.Emp_Name AS Manager_Name
FROM Employee e
LEFT JOIN Employee m
    ON e.Manager_ID = m.Emp_ID
ORDER BY e.Emp_Name;

```